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EU AI Act

AI Regulatory
Sandboxes:
A Breakdown

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AI sandboxes accelerate innovation while ensuring compliance. European countries use sandboxes to test high-risk AI in controlled environments, balancing experimentation with regulatory oversight.

EU Artificial Intelligence Act: AI Regulatory Sandbox Approaches: EU Member State Overview (May 2025)

Member State	Overview	Key Actors	Legal Framework	Operational Structure
	Planning phase for AI sandboxes; AI Service Desk provides guidance	Interdepartmental cooperation mentioned; authorities not specified	No framework published	Not defined
	No national sandbox; Flanders exploring one for SMEs/startups	EWI, Federal Data Protection Authority, market surveillance authorities, VLAIO, Digital Flanders, Knowledge Centre Data & Society	Federal framework extended w/ exemptions required for exemptions	Tiered services (basic, extended, application, testing, exit documentation)
	No sandbox; Sofia Tech Park as a testing lab	Unknown	Not adopted	Unknown
	Draft law to cover AI sandboxes	Not yet identified	Pending	Undetermined
	Plans to establish sandbox within 12 months of AI Act	Deputy Ministry of Research, Innovation, Digital Policy	Not established	Not defined
	Digital Regulatory Sandbox under National Recovery Plan; mainly fintech	CzechInvest, Ministry of Industry & Trade, Ministry of Finance	Status under AI Act unclear	Two phases: framework design, startup testing; tailored support; max funding ~6M€
	Sandbox launched 2024; GDPR-focused, expanding to AI Act	Datatilsynet, Digitaliseringsstyrelsen	GDPR compliance; no exemptions	Six-month program; legal guidance; project selection based on societal benefit, innovation
	Accelerate innovation removing regulatory barriers	Estonia: Ministry of Economic Affairs & Communications, other ministries	Not clear if under mandatory AI Act sandbox	Public calls, ministry collaboration, real-world validation, potential legislative follow-up
	Establishing national sandbox; EUSAIR pilot	Traficom, market surveillance authorities, Data Protection Ombudsman, EUSAIR pilot participants	Amendment to Act on Supervision of AI Systems; implementing acts pending	Applications to Traficom; consultation with authorities; details to follow EU guidance
	Data protection sandbox focused on AI in public services	CNIL, evaluation committees	GDPR compliance; no exemptions	Three phases: support, implementation, knowledge sharing; legal guidance only
	Institutionalized sandboxes; draft Regulatory Sandboxes Act (2024)	BMWK, Network of Regulatory Sandboxes	Experimentation clauses; draft act standardizes exemptions	Innovation Portal; formal mechanisms for experimentation, learning, and reporting; pilot up to 4 years



Member State	Overview	Key Actors	Legal Framework	Operational Structure
	No AI-specific sandbox	Unknown	Not established	Unknown
	Resolution adopted 2024; new enforcement body	Ministry for National Economy entity, Hungarian AI Council	High-level mandate; detailed legislation pending	Operational structure limited
	Mentioned in National AI Strategy 2024	Department of Enterprise, Enterprise Ireland, IDA, others	Not announced	Unknown
	<i>Sperimentazione</i> sandbox since 2020; pilot projects for AI	<i>Italia</i> Department for Digital Transformation, AgID	Law Decree 76/2020; temporary exemptions; certain sectors excluded	Applications with technical annexes; preliminary consultations; post-experiment reporting
	No sandbox yet; proposal for National AI Center	National AI Center; multi-stakeholder governance	Not defined	Not available
	Regulations approved to implement EU AI Act and AI sandbox	Innovation Agency, Communications Regulatory Authority, Ministry of Economy & Innovation	Amendments to Technology & Innovation and Information Society Services Laws	Operational details not specified
	<i>Sandkëscht</i> sandbox launched 2024; GDPR CNPD compliance		No legal exemptions; operates under GDPR; Three-stage process over 9–18 months; legal guidance only development	
	Plans for AI regulatory and data sandboxes	MDIA, Technology Regulation Advisory Committee, IDPC	Regulatory sandbox: temporary exemptions; data sandbox: GDPR for public benefit; monitored closely guidance	
	Proposal developed with multiple stakeholders	AP, RDI, sector-specific authorities	Guidance only; no legal exemptions	Three-pillar model; six phases from pre-registration to post-participation insights
	Implementing EU AI Act via Draft Act on AI Systems	Committee on Development & Safety of AI, Social Council on AI	Draft Act aligns with EU AI Act	Operational details not defined
	Strategy mentions sandboxes	Undecided	Not adopted	Not established
	Strategy mentions sandboxes	Undecided	No framework	Unknown
	Regional sandbox in Košice; focus on public administration	Innovation Center of Košice Region (ICKK)	Operates without national framework	Four-stage process: application, selection, implementation/testing, evaluation; best project may present internationally
	Developing legal and conceptual framework	Not specified	Under development	Undecided
	Sandbox launched 2022 under Royal Decree 817/2023; high-risk AI	AESIA, AEPD	Decree defines eligibility, selection, testing, compliance procedures	Projects implement AI Act requirements; authorities collaborate; final reports validate compliance





Member State	Overview	Key Actors	Legal Framework	Operational Structure
	Pilot for national sandbox; data protection sandbox (2022); focus on generative AI	IMY	GDPR and Swedish data protection; exemptions	Project-based; guidance on legal no questions; outcomes published; two projects completed, third underway

European Union member states are actively planning, piloting, or implementing AI regulatory sandboxes as part of their broader strategy to accelerate innovation while ensuring compliance with the EU AI Act. Countries like Austria, Bulgaria, Cyprus, and Ireland are in the early planning phase, defining strategic objectives and identifying potential actors, while others, including Denmark, Finland, Estonia, and Slovakia, have launched pilot programs to test AI solutions in controlled environments. These sandboxes provide structured frameworks for startups, SMEs, and public-private partnerships to navigate regulatory requirements, validate high-risk AI applications, and gain practical insights before broader market deployment.

Key actors across Europe typically include lead national authorities, sector-specific regulators, innovation agencies, and advisory boards offering ethical, technical, or policy guidance. Governance models range from centralized coordination in Germany and Finland to multi-stakeholder frameworks in Latvia and Malta, reflecting national priorities and sectoral considerations. By combining legal oversight, technical support, and collaborative evaluation, these initiatives create environments where experimentation is encouraged, while public safety, data protection, and compliance remain priorities.

Operationally, sandboxes adopt staged processes: eligibility assessment, application and selection, testing with regulatory guidance, and exit documentation validating compliance or informing legislation. While core AI innovation drives most value in sectors like software, fintech, and media, industrial and public-sector initiatives demonstrate that support functions—such as finance, compliance, and operations—are high-leverage areas. Strategically aligning sandbox design with sector-specific objectives ensures experimentation maximizes business and societal impact, delivering measurable innovation while mitigating legal, operational, and reputational risks.





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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

What Do We Do?

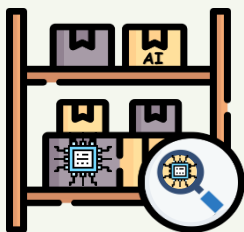
We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.



How Do We Do It?

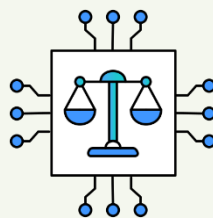
Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.



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Introduction

As the EU Artificial Intelligence Act (AI Act) progresses toward full implementation, AI regulatory sandboxes have become a vital tool for fostering safe, transparent, and innovative AI development. These controlled environments allow AI systems—especially those deemed high-risk—to undergo real-world testing, benefiting from regulatory oversight and flexibility during the process.

This report serves as a comprehensive guide for AI firms, regulators, and users engaged in designing and utilizing AI regulatory sandboxes. It synthesizes best practices, lessons from analogous frameworks (such as blockchain sandboxes), and actionable strategies to standardize sandbox operations across the EU. From classifying risks and ensuring legal interoperability to safeguarding user rights and monitoring AI systems post-market, the report explores opportunities to align sandbox activities with the AI Act's stipulations and broader digital governance objectives.

The introduction of regulatory sandboxes represents a significant step forward in the EU's efforts to establish a resilient and forward-thinking AI governance model. The initiative strikes a balance between fostering innovation and ensuring accountability under the AI Act, offering a structured environment to test AI systems while meeting regulatory standards. This fosters both innovation and public trust, encouraging the ethical and safe deployment of AI technologies.

The report also highlights the roles, responsibilities, and interactions among the key stakeholders involved in sandbox operations. It advocates for a collaborative approach that respects rights and supports AI experimentation. By engaging with sandboxes, firms can streamline compliance, regulators can enhance technical expertise, and users can benefit from more trustworthy, explainable AI systems before they hit the market.

Whether you are an emerging AI company testing a foundation model, a regulatory body refining enforcement strategies, or a civil society organization monitoring AI's societal impact, this report offers a structured framework for engaging with AI regulatory sandboxes. It sets the stage for responsible, scalable, and legally compliant innovation within the EU's AI ecosystem.

Best regards,

Sean Musch

Founder/CEO

AI & Partners



Key questions being asked about AI Regulatory Sandboxes





1. What is an AI regulatory sandbox?

An AI regulatory sandbox is a supervised, time-bound testing environment where companies can experiment with innovative AI technologies under the oversight of regulators. The goal is to foster safe innovation by enabling real-world testing while managing risk. Sandboxes also allow regulators to better understand emerging technologies and adapt legal frameworks accordingly. Participation offers developers and deployers a chance to refine compliance, improve product performance, and build trust with users and authorities—all before full-scale market launch. Sandboxes are increasingly used in sectors like fintech, healthcare, and transport across the EU and beyond.

2. Who can participate in an AI regulatory sandbox?

Participation is typically open to AI developers (providers), deployers (organizations applying AI), and sometimes academic or research institutions. The selection process favors entities with innovative, high-impact projects that benefit from regulatory guidance. SMEs and start-ups often receive priority to level the playing field. Applicants must demonstrate a clear use case, relevance to public interest or sector-specific goals, and readiness for testing in a controlled environment. Cross-border or multi-sector collaborations are also encouraged, especially in EU-coordinated initiatives. Participation is voluntary and not mandatory, but offers significant strategic, technical, and compliance-related benefits.

3. How do you apply to a sandbox program?

Application processes vary by country or regulatory body but generally include a formal proposal explaining the innovation, objectives, potential risks, and why sandbox participation is necessary. You may be asked to submit technical documentation, risk mitigation strategies, data management plans, and a timeline. In some cases, public sector sandboxes require alignment with broader goals, such as sustainability, public health, or digital transition. It's crucial to monitor calls for applications from relevant regulators and prepare materials in advance. Enterprises should also designate a sandbox liaison or cross-functional team to manage engagement throughout the process.

4. What are the main benefits of joining a sandbox?

Joining a sandbox offers significant advantages: (1) Real-time regulatory feedback that helps ensure early compliance; (2) Opportunity to test and refine products before wide-scale deployment, reducing legal and operational risks; (3) Improved credibility with clients, investors, and regulators; (4) Accelerated time-to-market by resolving uncertainties early; and (5) Contribution to shaping forward-looking, innovation-friendly regulations. It also fosters trust, as the public knows technologies have been vetted under supervision. In the EU, sandbox participation may also offer access to technical support, networks, or funding under programs like the Digital Europe initiative.

5. Are sandbox participants exempt from liability?

No. Under the proposed EU AI Act, participants in regulatory sandboxes remain fully liable for any harm caused during testing. Participation does not offer legal immunity or reduce obligations under existing laws. However, regulators may offer limited flexibility or tailored guidance to help navigate complex compliance challenges. Participants must implement robust risk management, follow applicable data protection rules, and ensure systems meet safety standards. The aim is to balance innovation with accountability. Companies should view sandboxes as collaborative spaces—not legal loopholes—designed to refine and responsibly test high-potential AI technologies.





6. What risks are involved in sandbox participation?

Despite benefits, participation involves reputational, operational, and legal risks. Disclosing proprietary systems in a regulatory environment may raise concerns about intellectual property exposure. Regulatory scrutiny could also reveal system weaknesses or unintended harms. If the sandbox involves real users or data, companies must take strict precautions to avoid breaches, discrimination, or safety issues. Failing sandbox criteria could delay market readiness. Additionally, organizations remain liable for any damage caused during testing. That said, these risks are generally outweighed by the opportunity to innovate responsibly, gain regulatory insight, and improve product robustness in a supervised environment.

7. Do sandboxes differ across EU Member States?

Yes, sandboxes can differ significantly by Member State in terms of governance, eligibility, scope, and rules. While the EU AI Act promotes coordination and harmonization, implementation is optional—so sandbox maturity and regulatory approaches may vary. This creates potential risks of fragmentation and “forum shopping,” where developers choose Member States with lenient rules. To address this, the EU is promoting multi-jurisdictional collaboration and knowledge sharing through a European AI Board. Enterprises should compare sandbox frameworks across jurisdictions to find the best fit for their product, but also advocate for clearer, standardized EU-wide participation models.

8. What does participation look like in practice?

Once accepted, participants enter a defined testing phase under regulator supervision. This may include regular check-ins, performance reporting, risk assessments, and system audits. Participants typically operate with some regulatory flexibility but must comply with baseline legal standards, especially around safety, fundamental rights, and data protection. Testing may use simulated data, restricted access, or limited user groups. At the end, results are reviewed to evaluate system compliance, performance, and readiness. Regulators may offer feedback, request changes, or approve system rollout. The process encourages collaboration, learning, and iteration between regulators and participants.

9. How are data protection and GDPR handled?

Data protection is a core concern. AI regulatory sandboxes must comply with the GDPR and ensure appropriate safeguards when using personal data. According to the draft EU AI Act, further processing of personal data in a sandbox is allowed only under strict public interest conditions (e.g., public health, safety, or environmental protection). National data protection authorities must be involved in the sandbox setup and oversight. Companies should implement privacy-by-design, conduct DPIAs (Data Protection Impact Assessments), and ensure transparency. Any exemptions or flexibility granted in sandboxes must not compromise the fundamental rights of individuals.

10. How do sandboxes support AI innovation in Europe?

Regulatory sandboxes are central to the EU’s innovation agenda, allowing for responsible AI experimentation while maintaining public trust. They help bridge the gap between cutting-edge technology and complex regulation by fostering collaboration between innovators and regulators. Sandboxes provide early clarity on compliance, reduce time-to-market, and support high-risk, high-reward projects. By integrating sandboxes into legislation, the EU aims to make regulation more agile and adaptable. For startups and SMEs, in particular, sandboxes offer a pathway to scale ethically and sustainably. Ultimately, they’re a strategic tool to balance innovation with risk management across Europe.

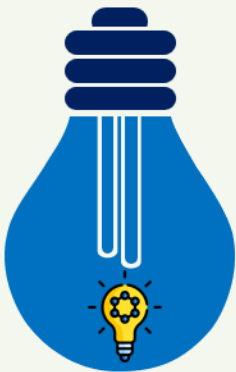




Key questions being asked about AI Regulatory Sandboxes



1. Introduction (Cohort Overview)



Why participate?

Participation allows organizations to innovate within a safe, supervised space. It helps build trust with regulators and stakeholders, minimizes compliance risks, and enables faster time-to-market for complex AI technologies. By joining, participants also influence regulatory understanding, helping shape practical and innovation-friendly rules that benefit the broader ecosystem.

How to participate?

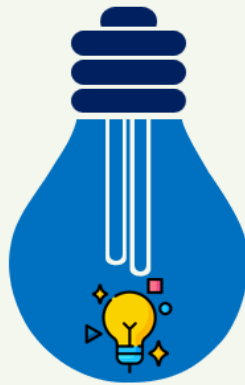
Entities typically apply through a national regulatory authority when a sandbox call opens. Applications usually require a business case, explanation of innovation, risk mitigation plans, and details on how the project aligns with regulatory goals. Priority may be given to high-impact, public-interest, or SME-led innovations.

What does participation involve?

Participation involves real-world testing of AI systems under regulator supervision, with agreed-upon constraints and oversight. Participants receive regulatory guidance but must also share data, cooperate with assessments, and comply with sandbox conditions. They remain liable for outcomes and may be required to exit if risks become too high.



2. Terminology, Laws, and Regulatory Framework



Why participate?

Businesses participate in sandboxes to gather operational insights, reduce market-entry risk, and improve product-market fit. Testing in realistic conditions uncovers usability issues, legal pitfalls, and unforeseen outcomes early—when they're easier and cheaper to fix. It also provides clarity on how to meet compliance standards.

How to participate?

To join, businesses propose a novel solution that benefits from real-world testing and regulatory engagement. Applications should demonstrate innovation, potential market value, and how the sandbox will support business development. Businesses must align with sandbox objectives such as safety, legal compliance, and responsible innovation.

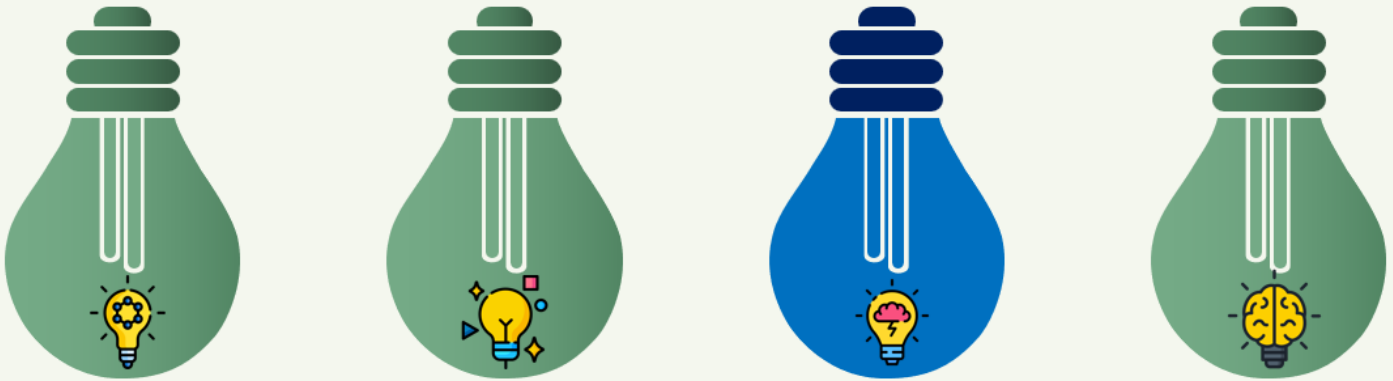
What does participation involve?

Participants engage in iterative testing, documentation, and dialogue with regulators. They monitor product behavior, collect performance data, and assess business outcomes. Teams adapt quickly based on feedback, refine compliance strategies, and often conduct risk assessments. It's a structured but agile cycle of innovation and validation.





3. Data Protection (GDPR)



Why participate?

Organizations participate to help shape smarter regulation by giving regulators direct insights into AI technologies in action. This ensures future laws are more practical, innovation-friendly, and less likely to create unintended burdens. It's a chance to collaborate, influence future compliance paths, and build long-term regulator relationships.

How to participate?

Applicants are selected based on how well their project contributes to regulatory knowledge. A strong application explains how the innovation challenges existing rules, raises novel compliance questions, or provides learning opportunities. Multistakeholder collaboration and transparency are often encouraged.

What does participation involve?

Participants must share documentation, facilitate regulator access to their systems, and engage in structured feedback loops. This could include technical walkthroughs, risk assessments, or joint workshops. Transparency, data-sharing, and active involvement in evaluation processes are central, as is a willingness to help regulators learn and adapt.



4. Relevance of eIDAS and Trust in AI Systems



Why participate?

Sandboxes let companies explore how AI performs in their specific sector—like health, finance, or energy—while navigating sector-specific rules. This boosts confidence that innovations are safe, relevant, and legally compliant. It also allows firms to tailor solutions to sector needs and build partnerships with key stakeholders.

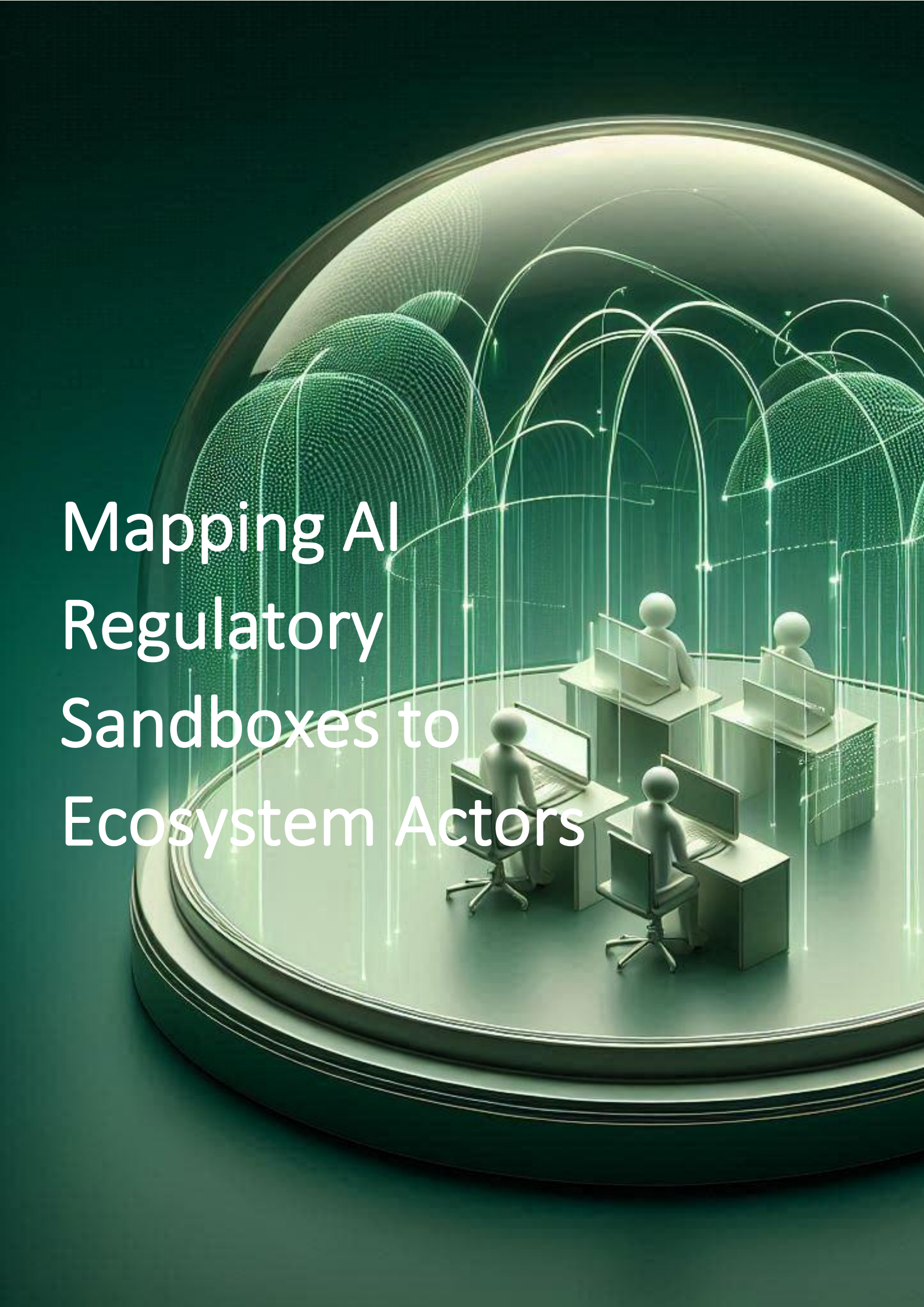
How to participate?

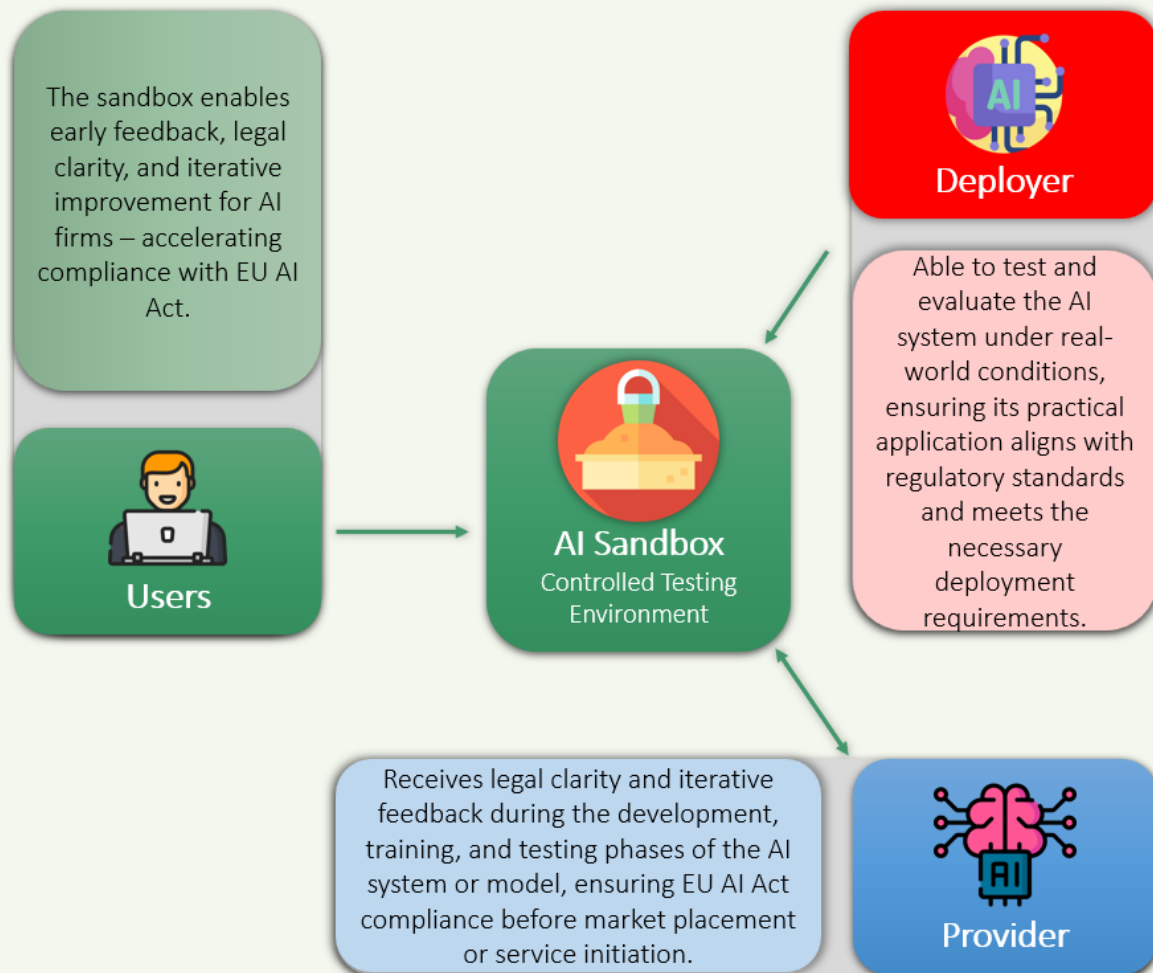
Interested parties apply to sandboxes tailored to their sector (e.g., fintech or health). Sectoral regulators typically open application windows outlining eligibility criteria. Applications must reflect sector relevance, novelty, and potential societal or economic impact. SMEs and public-interest projects often receive preference.

What does participation involve?

Participants conduct sector-specific testing, often with domain experts and compliance officers. They may use real-world data, simulate sector operations, and pilot technologies with limited user groups. Sector regulators provide tailored oversight, and participants adapt their systems to meet both general and domain-specific compliance frameworks.

Mapping AI Regulatory Sandboxes to Ecosystem Actors







AI Regulatory Sandbox		Actor		
Section	Description	Deployers	Providers	Users
Definition and Dual Purpose	Regulators observe new technologies firsthand through sandbox experiments. This improves their ability to craft flexible, informed policies that support innovation while protecting the public, balancing advancement with oversight in rapidly changing technological fields.	Deployers use regulatory sandboxes to better anticipate the rules and conditions they must follow when rolling out AI systems. Testing deployment strategies in a controlled environment allows them to spot operational, compliance, or ethical risks early. This proactive approach reduces costly errors and increases chances of successful scaling. Sandboxes provide deployers a crucial bridge between innovation and real-world application by exposing them to supervisory practices while adjustments are still feasible. Overall, the sandbox makes the transition to market smoother and more aligned with regulators' expectations, helping to avoid delays or penalties later.	For providers — those who design and create AI systems — regulatory sandboxes offer an opportunity to refine technical innovations while receiving early regulatory feedback. Providers can identify gaps in compliance, technical weaknesses, or unintended consequences without immediately facing strict penalties. Participating in a sandbox ensures that products are not only market-ready but legally robust, saving future redesign costs. Furthermore, sandbox involvement signals a commitment to responsible innovation, improving reputation with regulators and clients. It supports faster iteration and adaptation, giving providers an edge in highly competitive and fast-moving technological markets.	Although users do not directly participate in regulatory sandboxes, they benefit significantly from their existence. Products that emerge from sandbox environments are usually safer, more trustworthy, and better aligned with ethical and legal standards. Users can expect higher-quality, more reliable technologies that have been stress-tested under regulatory observation. Sandboxes thus contribute to greater consumer confidence in AI solutions, encouraging broader adoption. Additionally, by integrating user needs and feedback early in the development process through sandbox learning, providers and deployers can produce more user-centered innovations that better meet real-world needs and expectations.
Business Learning	Regulators observe new technologies firsthand through sandbox experiments. This improves their ability to craft flexible, informed policies that support innovation while protecting the public, balancing advancement with oversight in rapidly changing	Deployers utilize the sandbox environment to test operational strategies without full market risks. This enables them to uncover deployment challenges — such as user experience issues, cybersecurity vulnerabilities, or regulatory conflicts — in advance. Business learning in sandboxes allows deployers to refine how AI systems are	Providers leverage business learning in sandboxes to better align their AI products with market realities and legal expectations. Beyond technical development, providers gain insights into usability, scalability, and customer acceptance under actual conditions. Sandbox participation forces providers to adapt innovations to practical	For users, business learning in sandboxes means that the products they eventually encounter are more attuned to their real-world needs and concerns. Technologies are not just technically viable but are tested against practical business cases and user scenarios. Users can expect smoother, safer, and more intuitive interactions





	technological fields.	marketed, implemented, and scaled. It also supports adaptation to different regional or sectoral needs, depending on feedback received during the testing phase. Ultimately, deployers can fine-tune both their technology and deployment tactics, which increases the likelihood of a successful, compliant, and efficient market launch.	and regulatory constraints early on. It reduces costly post-launch compliance fixes and fosters a culture of continuous improvement. Providers also improve their ability to communicate the value and safety of their technology to customers, investors, and regulators, making it easier to secure market approvals and public trust.	with AI solutions born from sandbox experimentation. Furthermore, because business learning often involves iterative feedback loops, users indirectly influence product improvements, resulting in technologies that better reflect their expectations regarding functionality, privacy, fairness, and usability.
Regulatory Learning	Regulators observe new technologies firsthand through sandbox experiments. This improves their ability to craft flexible, informed policies that support innovation while protecting the public, balancing advancement with oversight in rapidly changing technological fields.	Deployers benefit from regulatory learning in sandboxes because they face fewer ambiguities when preparing for full deployment. Regulatory authorities use sandbox insights to refine enforcement guidelines and offer practical advice based on real-world examples. This helps deployers better understand how to meet compliance obligations without stifling innovation. Clearer, more predictable regulation reduces uncertainty, streamlining deployment processes and strategic planning. Moreover, deployers can engage more actively with regulators during sandbox participation, helping to shape pragmatic, business-friendly regulations that	Providers gain crucial insights from regulatory learning as it ensures that future rules and standards better reflect the realities of AI innovation. By participating in regulatory sandboxes, providers help inform the development of proportionate, innovation-friendly regulation. This proactive relationship with authorities reduces the likelihood of facing inflexible or outdated rules. It also empowers providers to align product development with forthcoming regulatory trends, ensuring smoother certifications, approvals, and market access. Engaging regulators early fosters trust, accelerates compliance readiness, and ultimately enables providers to innovate with more confidence and clarity.	Regulatory learning in sandboxes benefits users by ensuring that AI regulations are grounded in practical realities, leading to more effective consumer protections. Users can trust that regulatory oversight is informed by firsthand experience with emerging technologies, not theoretical assumptions. This leads to clearer rights, better safety measures, and higher ethical standards for the AI products users engage with daily. By improving regulatory frameworks through sandbox experience, the system better safeguards user interests, such as privacy, security, transparency, and fairness, without unnecessarily hindering technological progress.





		support safe innovation across industries.		
Sectors of Application	Sandboxes are now common in sectors like fintech (e.g., digital wallets), transport (e.g., autonomous vehicles), energy (e.g., smart meters), telecom (e.g., 5G deployment), and healthcare (e.g., disease prediction). They help tackle sector-specific innovation challenges under regulatory supervision.	Deployers working in sectors like fintech, healthcare, energy, transport, and telecom use sandboxes to trial AI deployments tailored to their industry's unique demands. Sandboxes allow them to address sector-specific risks (e.g., health data privacy, autonomous vehicle safety) before launching widely. Through supervised experiments, deployers can adapt AI systems to industry-specific regulations and market expectations. They can better predict barriers to adoption and fine-tune deployment strategies to sector nuances. Sandboxes, therefore, act as critical proving grounds, enabling deployers to validate both the technological soundness and the commercial viability of their sector-specific AI applications.	Providers developing sector-specific AI technologies gain substantial advantages through sandboxes by tailoring their solutions to the regulatory and operational needs of industries like finance, health, or telecom. Providers can test performance under sector conditions and gather valuable user feedback. Sandboxes ensure that innovations meet sector compliance standards, such as GDPR in healthcare or payment service directives in fintech. Early sector-based testing reduces time-to-market and boosts credibility with clients and regulators. Providers can also more easily identify opportunities for cross-sector applications by understanding how AI behaves under different operational frameworks and sector challenges.	Users in specific sectors (e.g., patients in healthcare, drivers using autonomous vehicles, fintech customers) benefit because sandboxes ensure that sector-specific AI products are rigorously tested before public release. Users can trust that AI solutions are optimized for their particular needs, whether ensuring financial transaction security or safeguarding personal medical data. Sandboxes also encourage the creation of more ethical, responsible technologies by making user impact a critical part of the testing process. Ultimately, users receive better-targeted, safer, and more reliable innovations aligned with both sector-specific regulations and their real-world use cases.



Calls to Action





Establish a Cross-Functional Sandbox Strategy Team

Create a dedicated team combining legal, technical, and business leads to explore and oversee sandbox participation. This team should assess how sandbox participation aligns with product roadmaps, legal readiness, and innovation goals. The objective is to ensure your organization is prepared to both contribute to and benefit from regulatory and business learning



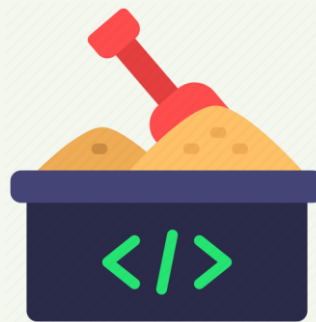
Pilot Real-World Use Cases in Sandbox Conditions

Identify AI products or features that could benefit from testing in a controlled environment. Focus on edge cases, compliance-sensitive models, or complex deployments. Use sandbox testing to refine models, gather user insights, and iterate on usability and market readiness while mitigating legal and reputational risks



Engage Regulators Early with Transparent Documentation

Prepare detailed documentation outlining your AI system's architecture, data flows, risk controls, and compliance measures. Share this proactively with sandbox regulators. Regular updates and open dialogue will help shape favorable, evidence-based regulation and reduce future compliance friction



Align Innovation with Sector-Specific Standards and Stakeholders

Map your AI solutions to sectoral regulations (e.g. GDPR in health, PSD2 in finance). Engage sector stakeholders—industry groups, end users, compliance bodies—to test assumptions and refine features. Sector-aligned sandbox participation helps tailor your product to real-world needs while boosting trust and adoption.

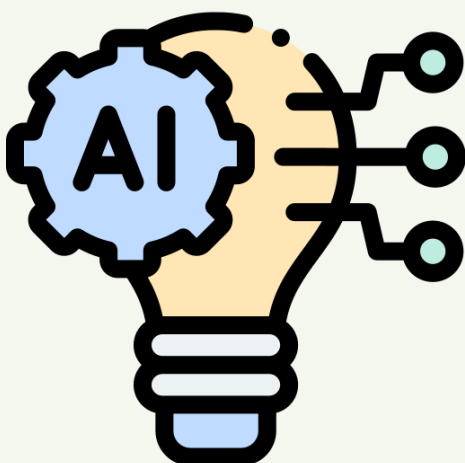




Conclusion

The introduction of AI regulatory sandboxes under the EU Artificial Intelligence Act (AI Act) marks a significant shift in how high-risk AI systems are developed, tested, and governed. This shift moves toward a more collaborative, transparent, and risk-based compliance model. By providing a supervised space for real-world testing, sandboxes help translate the AI Act's legal requirements into actionable steps, reducing uncertainty for innovators while ensuring robust user protections and regulatory accountability.

This report emphasizes how sandboxes act as a convergence point for AI firms, regulators, and users, facilitating early-stage feedback, technical oversight, and the creation of safer, more transparent AI systems. When designed effectively, these environments can accelerate alignment with the AI Act's standards on data governance, human oversight, and lifecycle risk management, fostering both trust and innovation across the EU's AI ecosystem.



However, the successful implementation of AI regulatory sandboxes requires more than just thoughtful policy design—it demands operational readiness and execution. Stakeholders need to adapt their internal compliance processes, documentation practices, and governance frameworks to meet the AI Act's expectations, especially for high-risk use cases defined under Annex III. Smaller developers and first-time participants may face steeper learning curves, highlighting the importance of early planning, regulatory guidance, and cross-functional collaboration.

Despite these challenges, forward-thinking organizations and authorities are demonstrating that AI sandboxes can streamline compliance, improve system quality, and shorten time to market. As AI continues to permeate diverse sectors—from healthcare and finance to public administration and civic infrastructure—regulatory sandboxes serve as a vital tool for developing AI systems that are innovative, lawful, ethical, and aligned with fundamental rights. In conclusion, successful outcomes in these sandboxes could influence future revisions of the AI Act and set a global standard for AI governance.





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Appendix A: Managing the Trade-off



Marijn Markus

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Compliance isn't a check-box exercise — it's a trade-off.

Working with AI means living in the space between privacy, transparency, and speed — and they rarely get along. Regulatory sandboxes make that tension visible: privacy rules clash with how data actually works, and transparency often exposes what we'd rather keep quiet. But it's that friction where AI projects truly shine. The challenge isn't just to control what AI systems can see, but to accept that uncertainty is part of the deal, and to explain this to folks who only think in Black or White. Because security without ethics is just a locked door — but ethics without realism is an open one.

Biography

Marijn combines technical expertise with a delivery of analysis, visuals and coaching, thanks to his experience and background in social science. As OSINT and sensor fan, Marijn leverages his affection for open data and internet culture to deliver unique insights and solutions. At Capgemini NL, Marijn has been leading AI efforts for over 9 years. From distinguishing science and fiction, creating friction and first introductions, To research, development, team management and implementing AI in production. Additional, Marijn is a frequent speaker at events on AI and data science. Most of all, Marijn seeks to improve people's lives using data.



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